Structure and Function of Macromolecules

LAB: INTRODUCTION TO *MAESTRO*

Report: No report.

Content

Introduction **1**

Setup **2**

Opening Maestro **2**

Importing a Molecule **2**

The Layout of Maestro **2**

Controlling the View **6**

The Workspace Navigator **7**

Visualising Proteins **8**

The Build Panel **11**

Building Molecules **14**

Saving Your Work **14**

Introduction

In the molecular docking labs, we will use the software Schrödinger Suite, which consists of several different applications that can be started from the interface Maestro. The Schrödinger software is primarily used for molecular modelling and computer based drug design and is an important tool for many research departments in the pharmaceutical industry. This exercise is a tutorial for the interface Maestro. It will be useful for later labs, which will instead focus on how the theory from the course can be used in practice

All labs will be performed in Linux environment on the computers in PC-rummet or Chipset. They can take several hours to complete, so don't forget to take breaks. When you do, please lock your computer (Ctrl+Alt+L).

Setup

Restart your computer. Start in Linux mode by pressing F12 repeatedly before Windows starts. Log in to your personal account. Open a terminal (click the button in the lower left corner of the desktop and then click System > Konsole). We will start by copying the files that are used in this practical. Write the following in the terminal:

```
cp -r /home/1MB412/Maestrolab/ .
```

(don't forget the dot) and change directory to the copied folder:

```
cd Maestrolab
```

Opening Maestro

Open Maestro by typing

```
export SCHRODINGER=/opt/schrodinger/

$SCHRODINGER/maestro
```

(The first command is only needed the first time you use Maestro.) The main window will open in a short while.

Importing a Molecule

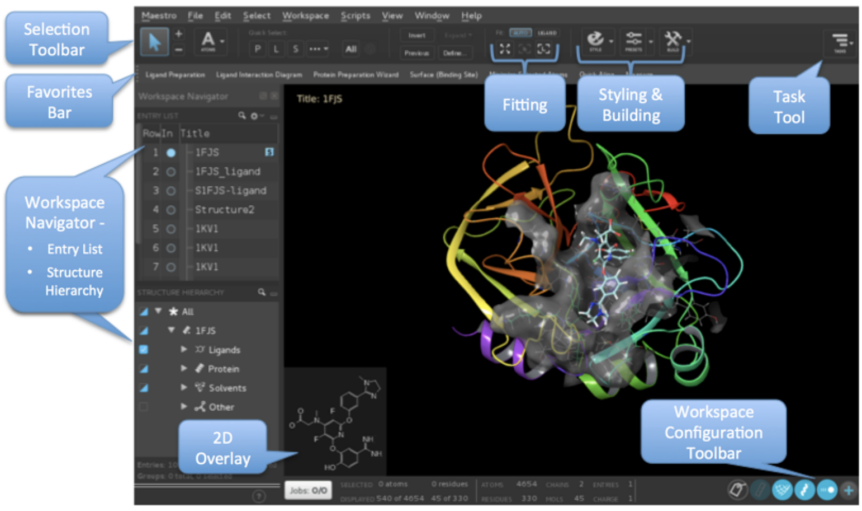
To get started using Maestro, you will import a structure into the Workspace.

1. Choose File > Import Structures from the File menu (or use Ctrl+I).
2. Open the file Donepezil.mae. Donepezil is an acetylcholinesterase inhibitor which is used as an anti-Alzheimer drug - you will learn more about this in the next sessions.

The Layout of Maestro

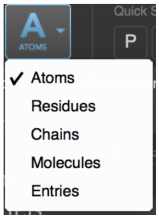
In the upcoming labs, you will be using the toolbars a lot, so it is useful to know a bit about the buttons, menus, and other controls beforehand.

The interface is shown below. Note that Donepezil should now be listed in the Workspace Navigator and the structure should appear in the Workspace.
The Selection Toolbar can be used to examine structures and interact with them. Click on an atom in Donepezil. The atom should now have a blue circle around it, indicating that it is selected. The Status Bar at the bottom should also tell you that 1 atom is selected. Hover your mouse over the different atoms in the molecule and see how the Status Bar changes. Shift-click to select multiple

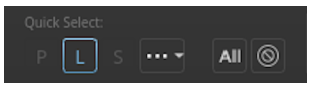


atoms or use the [+] and [-] buttons in the Selection Toolbar to add or subtract selections. As you do so, the number of selected atoms in the Status Bar should change accordingly.

By default, the Selection Toolbar will only pick one atom. Use the Pick Level menu button to choose a different substructure level. Your current pick level will be indicated by the letter next to your cursor in the Workspace.



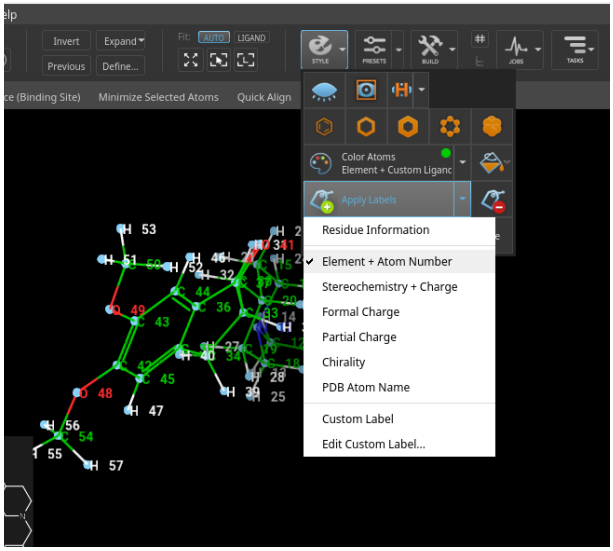
Common objects can be selected by using the buttons in the Quick Select area of the Selection Toolbar. Click **L** to select all ligand atoms in the Workspace and then click the Clear Selection button or click into an empty area of the Workspace to deselect the atoms.



You can use the Style Toolbox to modify the visual representation of your molecule. The Style Toolbox provides you with precise control over how structures are displayed in the Workspace.

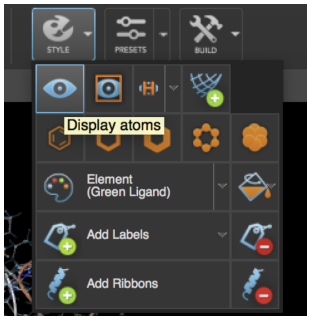
1. Select all atoms in the Workspace.
2. Click the arrow on the Style button to open the Style Toolbox. To display a label alongside the ligand with the elements and atom numbers of each

atom, select Apply Labels > Element + Atom Number. You will also see that the Annotations Toggle (in the bottom Workspace Configuration Toolbar) is now lit.



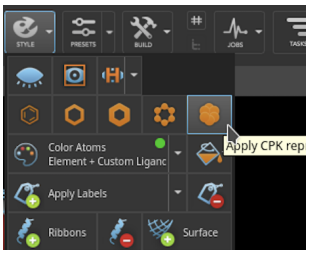
3. Delete the labels from selected atoms by clicking on or use the Annotations Toggle to hide the labels.

4. The visibility buttons in the Style Toolbox allow you to hide and show selected atoms in the Workspace. With all atoms selected, click on the Display/Undisplay visibility toggle eyeball to hide Donepezil. Clicking multiple times will alternate between hiding and showing selected atoms.

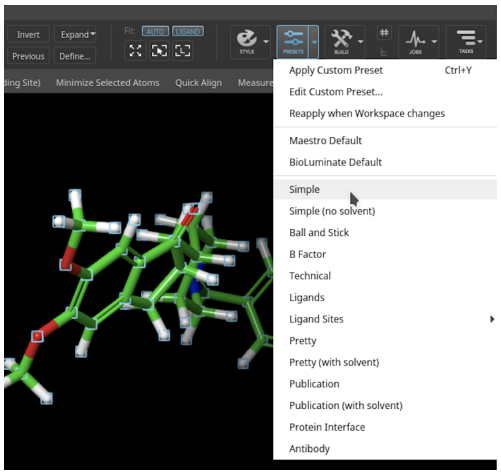


5. You can also select a small part of the molecule using your mouse pointer and display only selected atoms (the cropped eyeball icon). Part of the molecule should no longer be shown in the Workspace. This is useful, for example, if you only want to look at the binding site + ligand atoms (using the "..." button in the Quick Select area).

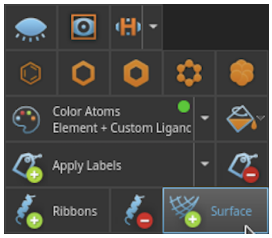
6. Several molecular renderings are available in Maestro: wireframe, thin tubes, thick tubes, ball-and-stick, and CPK. Select all atoms and change the representation of the atoms and bonds to CPK and see how the Workspace changes. Afterwards, switch back to wire representation.



7. Maestro also has a variety of Preset styles for specific visualisations of the Workspace. These are called one-click styling since they apply to all atoms in the Workspace, regardless of visibility or selection. Try some of the presets and then change back to Maestro Default.



8. Create a molecular surface on selected atoms by using the Style Toolbox. Hover your mouse over the surface until the symbol next to your cursor changes to a grid. Now right-click and select Display Options to adjust the style of the surface. You can use the Color scheme controls here to color the surface according to the values of the chosen property. Delete the surface by right-clicking and selecting Delete or by using the Surface Toggle in the Workspace Configuration Toolbar.



Controlling the View

It is possible to automatically fit the structure to the Workspace using the Fit buttons in the Selection Toolbar. Try to fit the view to all visible atoms in the Workspace.



The Workspace view can be changed by using the View menu (in the top panel) or by using the mouse. We will focus mainly on the mouse controls below.

Zooming in or out:
Place the pointer anywhere in the Workspace and use the Middle+right button on the mouse or the scroll wheel.

Translating the molecule:
Hold down the right mouse button and move the mouse.

Rotating the molecule:
Hold down the middle mouse button or scroll wheel and move the mouse. This will rotate the structure in the x and/or y direction (depending on how you move the mouse).

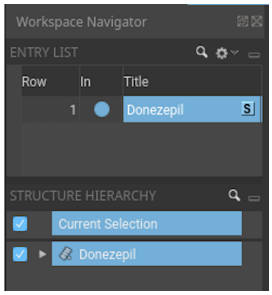
If you want to make sure you rotate only in the x direction or the y direction, hold down the Shift key along with the middle mouse button and move in that direction. To rotate the molecule around the z axis, hold down the Ctrl key instead of Shift.

Saving the view:
You can save the camera view and return to it later using the View menu.

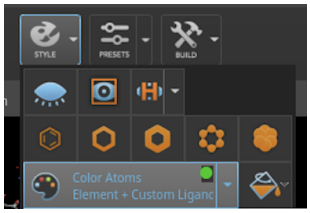
1. Click View > Save Camera View. You can use the default name in the Save View dialog box that appears and click OK.
2. Rotate the structure in any direction and save this new camera view in the same way as above.
3. Now return to the view you first saved by using Restore Camera View.
4. Manage Camera Views allows you to play the different views as a movie.

The Workspace Navigator

The Workspace Navigator (left column) is where Maestro stores structures that you have imported.

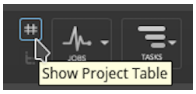


1. Import a second molecule with the name Galantamine.mae.
2. You can see all of your imported molecules under the Entry List. This section lets you control which entries are included in the Workspace and whether they can be edited. Both Donezepil and Galantamine should now be listed, but only the latest import is shown in the Workspace.
3. Change the colour of Galantamine to something else, so that you can more easily distinguish between the two ligands.

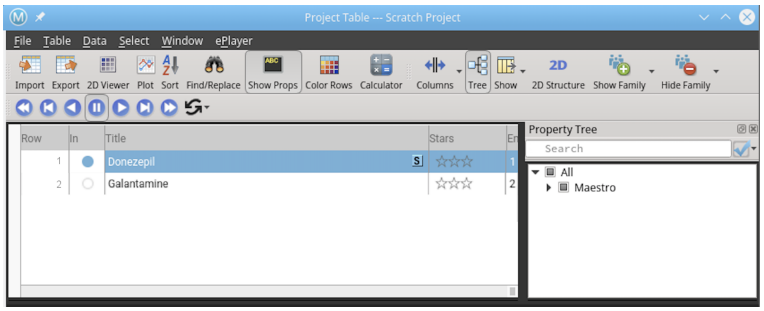


4. You can choose which entries are visible in the Workspace by toggling the circle in the "In" column next to the entry title. Click on the empty circle next to Donezepil to include that structure in the Workspace. Galantamine will now be excluded.
5. To include both entries in the Workspace, Ctrl-click on the circle next to Galantamine. Both structures should now be shown.
6. Ctrl-click a second time to remove Galantamine from the Workspace. Note: If you want to remove all structures from the Workspace, the fastest way is to right-click on the Workspace and choose "Clear Workspace".
7. To include a range of entries in the Workspace, Shift-click on two molecules entries. This will also include all entries in between the two you selected.
8. To create a new entry of Donepezil in the Workspace, right-click on its name and select Duplicate > In Place. You should now have two entries of Donepezil in the Workspace Navigator. This can be useful if you want to keep the original structure and modify a copy of it using Builder - more on this later.

9. Delete the second entry by right-clicking and clicking Delete.
10. Now open the Project Table (top right).



You will find your imported molecules listed in the table.

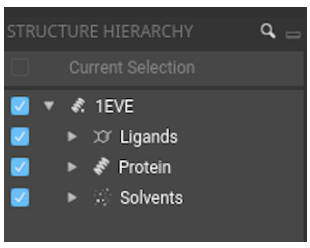


11. Let's calculate the molecular weight through the Project Table. Choose Data > Add Property > Standard Molecular Property. A list of possible properties is now shown. Choose Molecular weight and calculate for All Entries. You will see that a new column has appeared at the far right of your Project Table, with the molecular weight of each molecule.
12. We are now done with these structures, so close the "Calculate Properties" panel and delete all entries from the Project Table or from the Workspace Navigator.

Visualising Proteins

Looking at biomolecules can be difficult since they are large and contain many atoms. In Maestro you can simplify the view in many ways to make it easier.

1. Import the structure 1EVE.pdb. This is the x-ray crystal structure of Donepezil bound to the enzyme acetylcholinesterase.
2. You might have already noticed that the Workspace Navigator also contains a section entitled Structure Hierarchy.



This shows a list of the contents of all the entries that are included in the Workspace, classified into components of various types (shown in a tree).

You can access this by clicking the arrow next to 1EVE. If you place your pointer over an entry in the tree, a plus sign will appear to the right. Clicking will open up a Style Toolbox for this individual entry, allowing you to style it regardless of what is selected in the Workspace. Click once on Ligands to select the corresponding atoms in the Workspace.

3. Double-clicking on Ligands will select and zoom the ligand into view.

4. Click on the visibility check box (blue square) at the left edge of the "Solvents" row to hide the solvent atoms in the Workspace. Note that the visibility check box is partially coloured if only some of the atoms are shown and not coloured at all if all atoms are hidden.

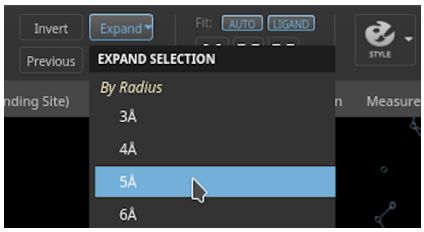
5. Click on the arrow next to "Protein" to expand the folder and then click on the arrow next to Chain A.

6. Double-click on any of the residues to zoom in on this part of the protein instead.

7. Let's focus a bit more on the ligand. Display only the ligand by Quick Selecting the ligand (only) and adjust the view to Fit the ligand to the Workspace.

8. Use the Style Toolbox to Display only selected atoms and change the representation of the atoms and bonds to ball-and-stick.

9. When looking at proteins, it is often easier to look at a small part. You can look at all residues within a certain distance from the ligand by Expanding the selection. Expand it by 5 Å and display the selected atoms.



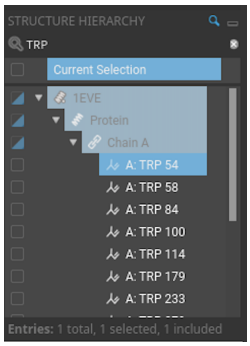
10. Remove the ligand from the Workspace selection by clicking the [-] button in the Selection Toolbar and then clicking on [L] under Quick Select. Now apply a label with Residue Information. Which amino acids do you find in the ligand binding site?

11. It is often important to get a picture of the secondary structure of the protein. Click "P" in the Quick Select area to select the protein and then on the "Ribbons" button in the Style Toolbox. Zoom out to see the alpha helices and beta strands more clearly.

12. Delete the ribbon representation or temporarily hide it using the Ribbons Toggle .

13. Select and view only the ligand again. Let's also display the tryptophan amino acids in the Workspace. To locate specific items in the Workspace, we can use the Search icon of the Structure Hierarchy. Type "TRP" into the field. All tryptophan residues are now listed under Chain A.

14. Shift-click on the first and last entry to select the entire range.

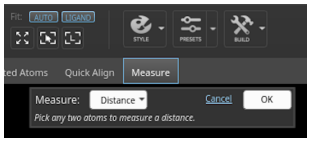


15. Display all selected atoms. Notice where in the protein structure the tryptophans are situated.

16. Close the search field and double-click on Ligands in the Structure Hierarchy.

17. Let's now do some measurements of the ligand. Apply a label to each ligand atom with its PDB atom name using the Style Toolbox.

18. You can use the Measure button on the Favorites bar to check distances and angles in the Workspace. Click Measure; a banner will appear.



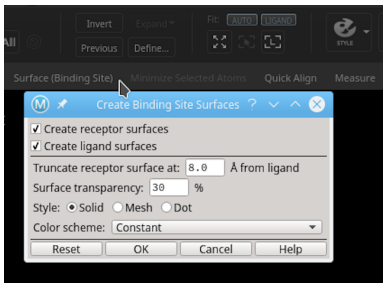
Pick ligand atoms C28 and C21. This will give you the distance in Ångström. Change the measurement option to angle and pick C15, C16 and C11 to get the angle in degrees. Change the option in the banner to dihedral and pick C11, C12, C13 and N14 to get the torsion angle.

19. Click the OK button on the banner to exit this mode. You can review measurements using the Measurements toggle (through the "..." icon above).



Use the toggle to delete all measurements and hide all labels.

20. It can be useful to add a surface to the cavity of the ligand binding site. This is easily created by the Surface (Binding Site) button. Quick Select Binding Sites + Ligands (using the "..." button) and create receptor and ligand surfaces using the default settings.



21. You can change the transparency, style, and colour of your surfaces by clicking on the Surface button.

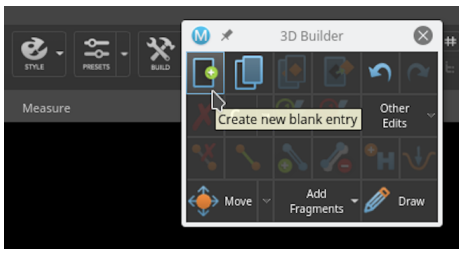
22. Clear the Workspace when you're done.

The Build Panel

In Maestro you can build your own molecules or modify existing ones. This is done with the help of the 3D Builder palette (the Build toolbox). You will be using this tool to design your own drugs in the upcoming lab, but first, you will become familiar with how to perform some basic tasks.

Note that if you make a mistake, you can select "Undo the last action" in Builder (or use Ctrl+Z).

1. Create a new entry and choose a title.



   This entry will now appear in your Workspace Navigator.

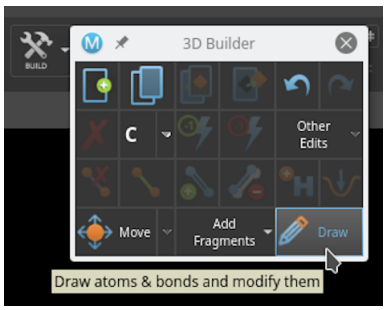
2. Click on Add Fragments in the Build Toolbox and add cyclohexane to your Workspace.

3. Select one of the hydrogens in cyclohexane and click on the hydroxyl fragment (HOH) in Builder to get an alcohol.

4. You can also delete atoms and bonds. Right-click on one of the bonds between two carbons in your molecule and choose "Delete Bond". Your ring should now be broken.

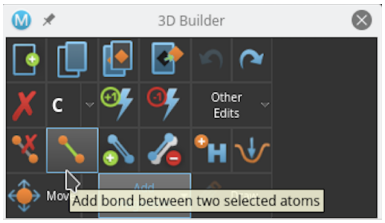
5. Right-click on one of the hydrogens in your molecule and choose "Delete".

6. You can also draw structures on free hand by using the Draw button of the Build toolbar.



7. Select Draw and use your mouse to draw a single line in the Workspace to create ethane.

8. Click and drag the pointer between one of the carbon atoms of ethane and a carbon on your ring-opened structure. This will add the ethyl substituent to your structure. Alternatively, you can also click "Select by picking" (to leave the drawing mode) and select the two carbon atoms and then add a bond between the atoms.



9. By default, you will draw carbon atoms. It is also possible to draw other elements or to change the atoms in a molecule.

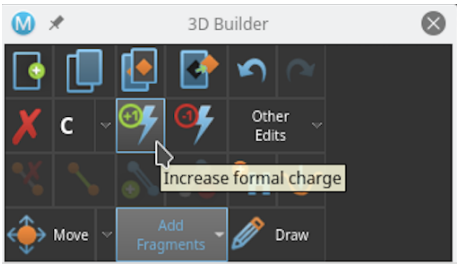


   Choose Cl from the Draw Structure menu on the Build toolbar. Click on one of the hydrogens in your molecule to add a chloride substituent.

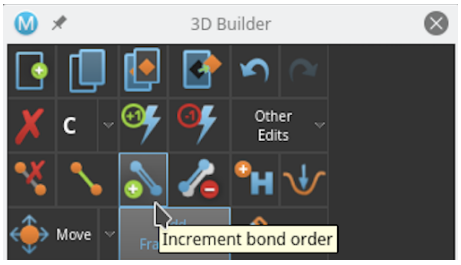
10. Grow your molecule by adding an amine group. Choose N and simply click on one a hydrogen atom on one of the ends of your molecule.

11. Next, you will learn how to fuse your molecule with a ring structure. Select one of the bonds in your molecule and then click on the cyclopentane button on the Fragments toolbar.

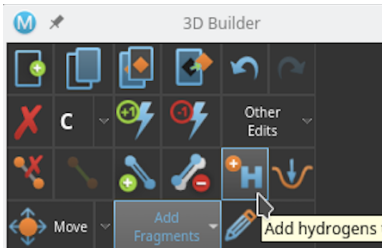
12. You can change the formal charge by using the buttons in the Build toolbar. Click on the nitrogen in your structure and select "Increase formal charge".



13. You can also change the bond order in the Build toolbar. Select a bond in your molecule and click on "Increment bond order".

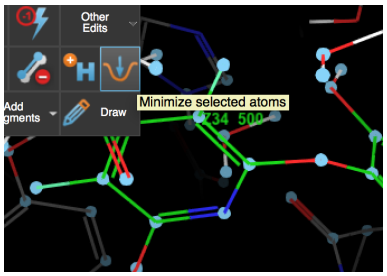


14. The molecule needs hydrogens to be complete. Maestro allows you to automatically add hydrogen atoms wherever they are needed. Select all atoms and then click "Add hydrogens to selected atoms" to build the necessary hydrogen atoms.



15. Your molecule is now complete, but if you rotate it, you will see that the structure still looks somewhat like a ring. Use the Move button menu in the Builder to rotate dihedrals and change lengths and angles in your molecule.

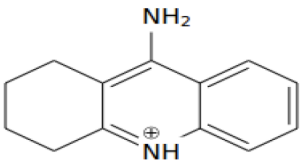
16. Clean up the geometry by selecting all atoms and clicking on "Minimize selected atoms". This will perform a quick minimization on all selected structures in the Workspace.



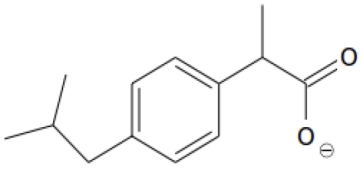
17. Maestro also has the option to draw structures in the 2D Sketcher. Click on Edit > 2D Sketcher and use it to modify the structure (or create a new one) that you then transfer to 3D by either Updating Entry or Saving as New.

Building Molecules

Build the following structures, one at a time. Export each structure as .mol2-files (see below).



Tacrine



Ibuprofene

Saving Your Work

In Maestro, you can save a structure in many different file formats. .pdb is the format used by the Protein Data Bank (www.pdb.org) that you might have come in contact with before. Maestro also has its own file format .mae.

1. Click on File > Export Structures from the menu bar. You can also right-click on an entry title in the Workspace Navigator. Only the structure information is saved with this option.

2. You can save the project you are working with, with all of your current settings (camera view, atom colours etc.), by using the "Save Project As" button from the File menu.

The practical ends here, but if you have time left you are welcome to stay and play with the software.

Please inform the teachers before you leave.